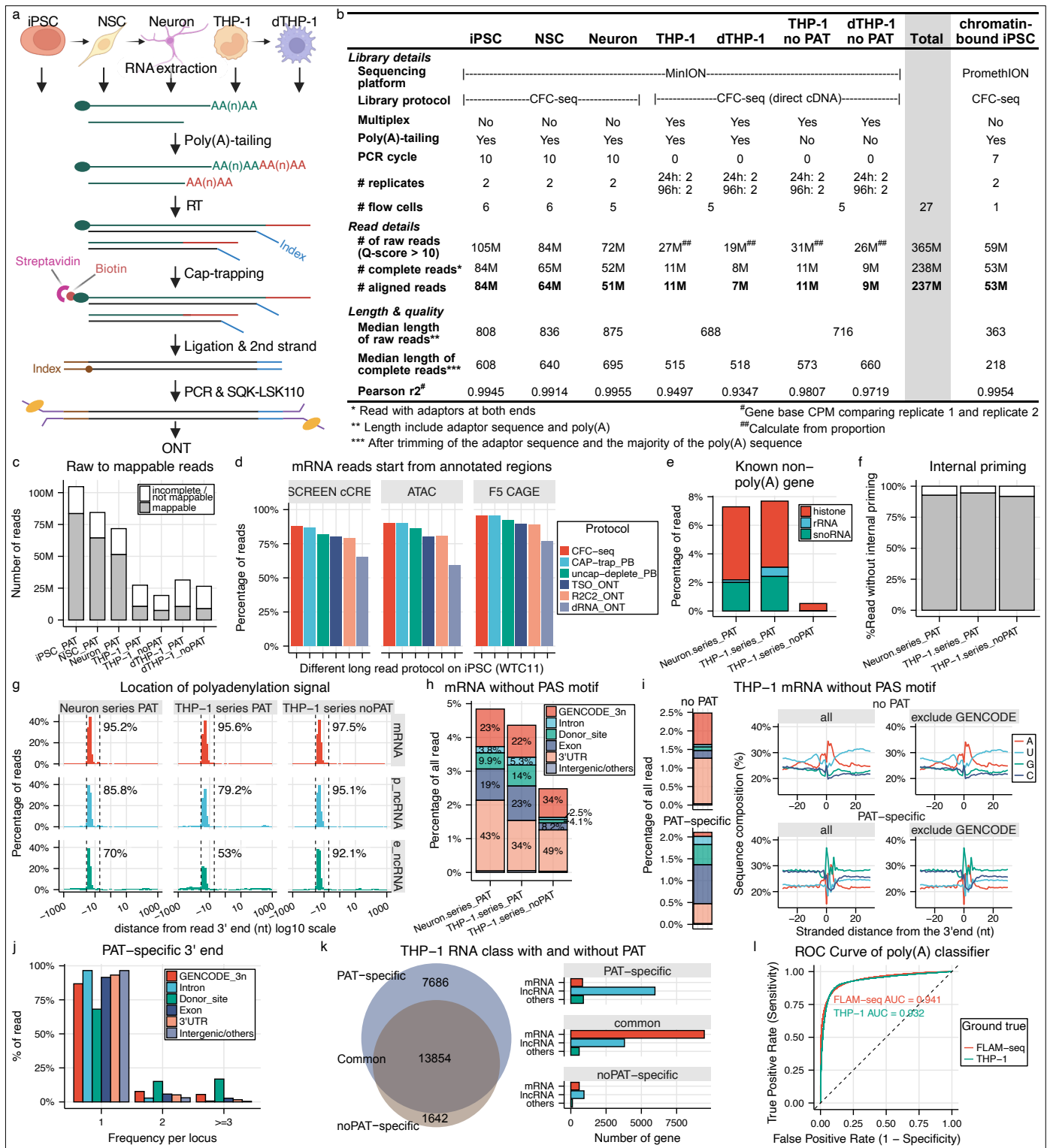




Extended Data Figure 1 | Technical validation and read statistics of the CFC-seq platform.

a, CFC-seq experimental workflow. **b**, Sequencing statistics and library metrics for all generated datasets. **c**, Quality control breakdown of long reads (base-calling score ≥ 10) into incomplete (missing adaptors), non-mappable, and mappable fractions. **d**, Intersection of mRNA 5' ends with SCREEN cCREs, ATAC-seq peaks, and FANTOM5 CAGE clusters. Long-read datasets from iPSC WTC11 were obtained from ENCODE LRGASP (Pardo-Palacios et al. 2024). **e**, Enrichment of non-polyadenylated (non-poly(A)) transcripts in libraries following in vitro poly(A)-tailing (PAT). **f**, Evaluation of internal priming based on genomic sequence composition downstream of read 3' ends. **g**, Polyadenylation signal (PAS) motif frequency (motif score > 3 , -35 to -5 bp upstream of the 3' end) in reads across mRNA, promoter-ncRNA, and eRNA classes, comparing $\pm$ PAT libraries. **h**, Genomic distribution of non-PAS-associated mRNA 3' ends to explore fraction of potential nascent RNA. **i**, Characterization of emergent mRNA 3' ends lacking PAS motifs in THP-1 $\pm$ PAT libraries. Left: genomic localization relative to coding exons and introns. Right: nucleotide composition frequency surrounding 3' end sites (± 30 nt). **j**, Frequency of PAT-specific mRNA 3' ends across genomic features, showing accumulation at splice donor sites. **k**, Overlap of genes detected by CFC-seq $\pm$ PAT and the distribution of their RNA classes. **l**, ROC curve for the sequence-based poly(A) classifier, validated against FLAM-seq 3' ends (as positives) and PAT-specific 3' ends (as negatives).